

EARNINGS

ENIGMA:

CRACKING

THE CODE

WITH K-NN

AND NAÏVE

BAYES

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CAPSTONE PROJECT IN BUSINESS ANALYTICS (BA-64099-024)

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1. INTRODUCTION

1.1. Abstract – Project Goal

The project aims to develop a machine learning prediction model that can accurately predict if an individual earns less than or more than \$50,000.00 annually. Which can be based on various features or variables such as socio-economic and demographic factors. Currently knowing what factors influence an individual's income can lead to insights into economic trends and consumer behavior. The variables in our dataset include age, marital status, education level, occupation, relationship, race, sex, hours worked per week, and country of origin.

The propositions of this project are mega. One such use can be in the form of marketing in the advertising industry. The companies can use this model to select what advertisement to target to which customers based on their income to increase their sales. For example, a superstore company can advertise a premium product for the people whose income are higher than 50K and a value product for the individuals who earn less than 50K. Another use for this model can be for policy-making, financial services, and any social programs that will impact the public effectively. This can be done by having a deeper understanding and predicting an individual's income.

There are similar projects that have been done but this case scenario is done by using KNN and Naïve Bayes prediction model with open-source coding software 'R'. By doing this project in R we want this analysis to be widespread for those who want this analysis solution for their business on a budget-friendly and similar basis. Our focus of this project is to simplify the whole process as simple as possible and very practically accessible by ensuring the variables we use are very easy to collect and understand. Which will enable an individual or a small organization to conduct their own research or model prediction without incurring major costs.

In short while using statistical techniques, machine learning, and open-source software this project wants to build a predictive model that not only predicts with accuracy but also cost-effectively. Thus, helping a budget-friendly individual or an organization to make data-driven decisions or advertising to get an edge at a fraction of the cost.

1.2. Problem Statement – Background – Design

To tackle the project, we have primary goals which are like business analytics life cycle to help with the creation of the project. The primary goals are Understanding the data, Factors influencing the income, Predictive model – Choosing an algorithm, Evaluating the model's performance, and addressing any ethical concerns.

1.2.1 Understanding the data

Make sure the dataset is ideal for model preparation, where the data consists of required data points and outcomes which can help the model to train. Limited missing data accurately represents the population, where the data is not skewed based on one demographic or one type of income level, which will result in training the model inaccurately.

1.2.2 Factors influencing the data

Accurately finding the correlation of factors that are directly related to income, which the model can concentrate more on to get the prediction more accurately. This is mainly done to gain insights into variables that are strong predictors of the outcome.

1.2.3 Predictive Model (Choosing an Algorithm)

Building a model that can accurately predict based on the factors provided if the income of an individual falls above or below 50k. The algorithms that will be used are KNN classification and Naïve Bayes classification.

1.2.4 Evaluate the Model's Performance

According to the no free lunch theory, different data sets on different algorithms work differently, even if one algorithm is superior in the methods it uses it's always better to check the accuracy of those algorithms against each other with the same dataset. (Brownlee, No Free Lunch Theorem for Machine Learning, 2021)

1.2.5 Ethical Concerns

Addressing any biases in the model or wrongly assuming one type of variable results in higher income. We will need to troubleshoot any such issues if necessary to ensure fair and ethical use of prediction. (Writer, 2024)

2. DATA EXPLORATION AND CLEANING

2.1. Data Overview

The data set is taken from Kaggle, the data set consists of 15 variables. 14 will be supporting variables and 1 will be an output variable. The 14 variables consist of 6 quantitative variables, 1 binary variable, and 7 qualitative variables. The data consists of 32,561 rows and 15 columns. The output column is 'Income', which is a binary classification.

2.2. Data Cleaning

The data has missing values of 0.87% of the total data. Since this is below the 5% limit, we will be deleting these lines. The missing data are denoted by '?'. These values are replaced by 'NA'. This is done to handle these values better in R while processing the data. A few numerical variable formats were formatted as a character which had to be converted to integer or as numeric.

2.3. Descriptive Statistics

Descriptive statistics provides concise summaries of a dataset's characteristics, which helps in understanding the dataset by offering brief overviews and measurements of its features. For this project, we used 3 methods to perform the descriptive statistics.

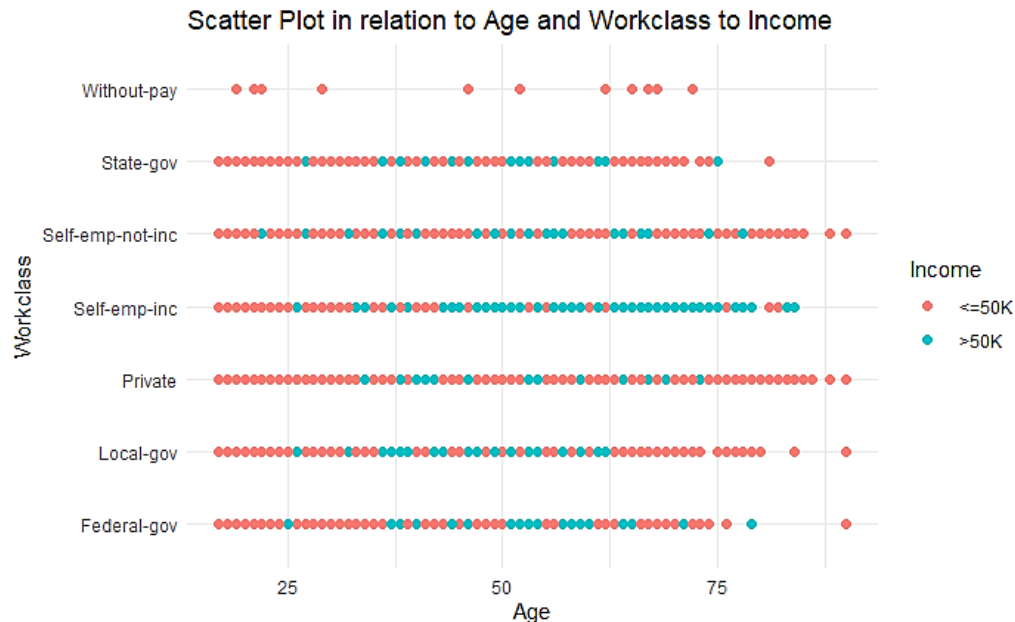
2.3.1. Summary Statistics

These statistics calculate the measures of central tendency, dispersion, and shape of a dataset. The statistics include mean, median, standard deviation, min-max, and inter quartiles. This not only shows the statistics but also the character type of each variable based on which we can convert the data type so it's easier for R to understand the data.

Age	workclass	Final.weight	Education	EducationNum	Marital.status
Min. :17.00	Length:30162	Min. : 13769	Length:30162	Min. : 1.00	Length:30162
1st Qu.:28.00	Class :character	1st Qu.: 117627	Class :character	1st Qu.: 9.00	Class :character
Median :37.00	Mode :character	Median : 178425	Mode :character	Median :10.00	Mode :character
Mean :38.44		Mean : 189794		Mean :10.12	
3rd Qu.:47.00		3rd Qu.: 237629		3rd Qu.:13.00	
Max. :90.00		Max. :1484705		Max. :16.00	
Occupation	Relationship	Race	Gender	Capital.gain	capital.loss
Length:30162	Length:30162	Length:30162	Length:30162	Min. : 0	Min. : 0.00
Class :character	Class :character	Class :character	Class :character	1st Qu.: 0	1st Qu.: 0.00
Mode :character	Mode :character	Mode :character	Mode :character	Median : 0	Median : 0.00
				Mean : 1092	Mean : 88.37
				3rd Qu.: 0	3rd Qu.: 0.00
				Max. :99999	Max. :4356.00
Hours.per.Week	Native.country	Income			
Min. : 1.00	Length:30162	Length:30162			
1st Qu.:40.00	Class :character	Class :character			
Median :40.00	Mode :character	Mode :character			
Mean :40.93					
3rd Qu.:45.00					
Max. :99.00					

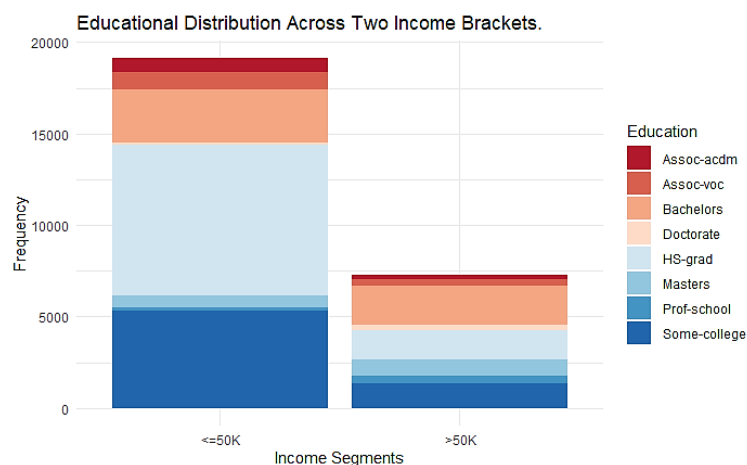
2.3.2. Scatter Plot

This scatter plot shows that most of the working-class population earns less than 50K which is indicated by the red dots except for the 'Self Employed' working class which leans towards the more than 50K income. As a whole less than 50K incomes are most common in certain working classes they exhibit a more balanced distribution of income levels.



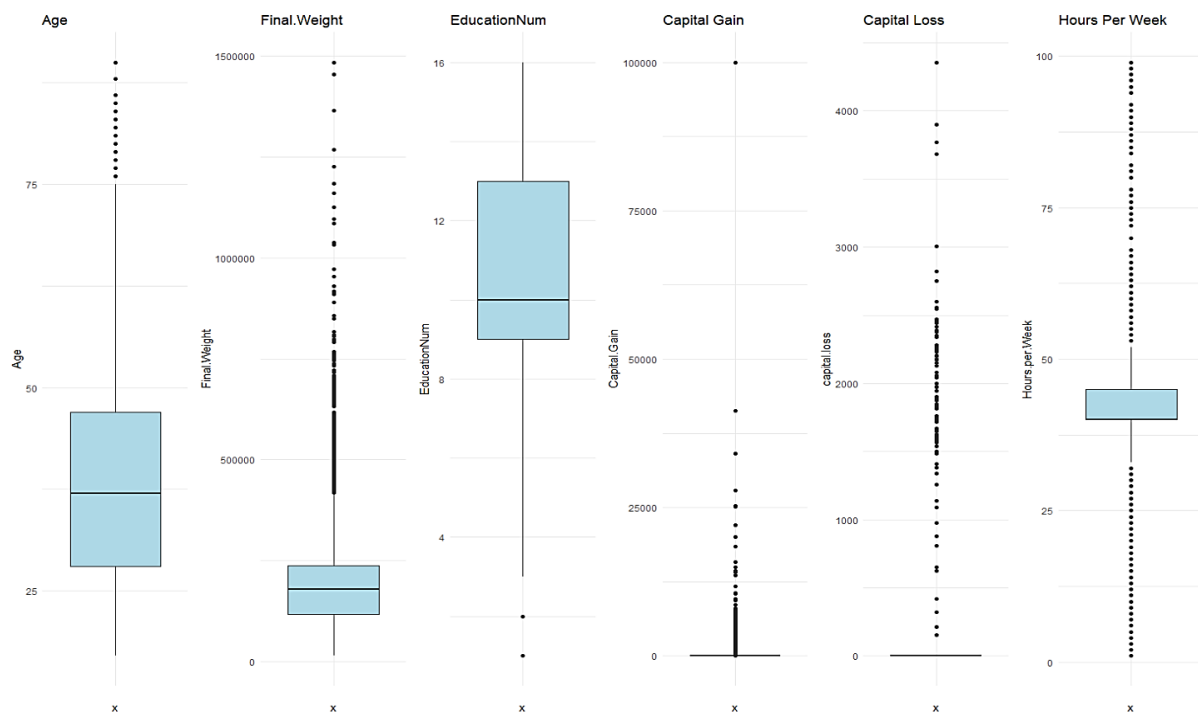
2.3.3. Bar Plot

Based on the bar plot, it is evident that less than half of the population earns more than 50K. The stacked bar graph shows that education does not have a significant impact on a person's income, as the distribution of education levels in both income scenarios is similar, except for high school graduates. It appears that a higher percentage of high school graduates fall into the less than 50K income category. Therefore, based on the stacked bar analysis, education does not seem to play a crucial role in income distribution.



2.3.4. Box Plot

The investigation reveals that some variables contain outliers. These outliers indicate that a small percentage of the data points in the dataset deviate significantly from the average range of the dataset. It's important to note that these values are genuine data points that could be highly significant in the context of the study, rather than indications of recording errors. It's crucial to understand that these numbers are not incorrect entries, even when outliers are present. Instead, they may represent unusual or special cases within the dataset. Identifying and analyzing outliers can offer valuable insights into the underlying trends and behaviors that the data aims to capture.



3. DATA TRANSFORMATION

In this step, we convert and handle the data so that the model works as assumed and this step ensures the easy reproducibility of the model. Below are the data transformation processes done.

3.1. One Hot Encoding

In the overview section of our report, it was noted that our dataset contained both categorical and numerical variables. To handle the categorical variables, we opted to convert them into dummy variables using the 'One-Hot-Encoding' technique. This is mainly done so that the model can read and understand the categorical value and its importance in a numerical sense. (Brownlee, Why One-Hot Encode Data in Machine Learning?, 2020)

The variables that were converted into one hot encoded data are work class, education, marital status, occupation, relationship, gender, native country and income.

3.2. Normalization

Normalization is a method where all the variable scales are converted into one so that the model can know the importance of each variable without the complexity of different scales. This ensures that the model performs accurately, consistently, and without biases. (Mulani, 2022)

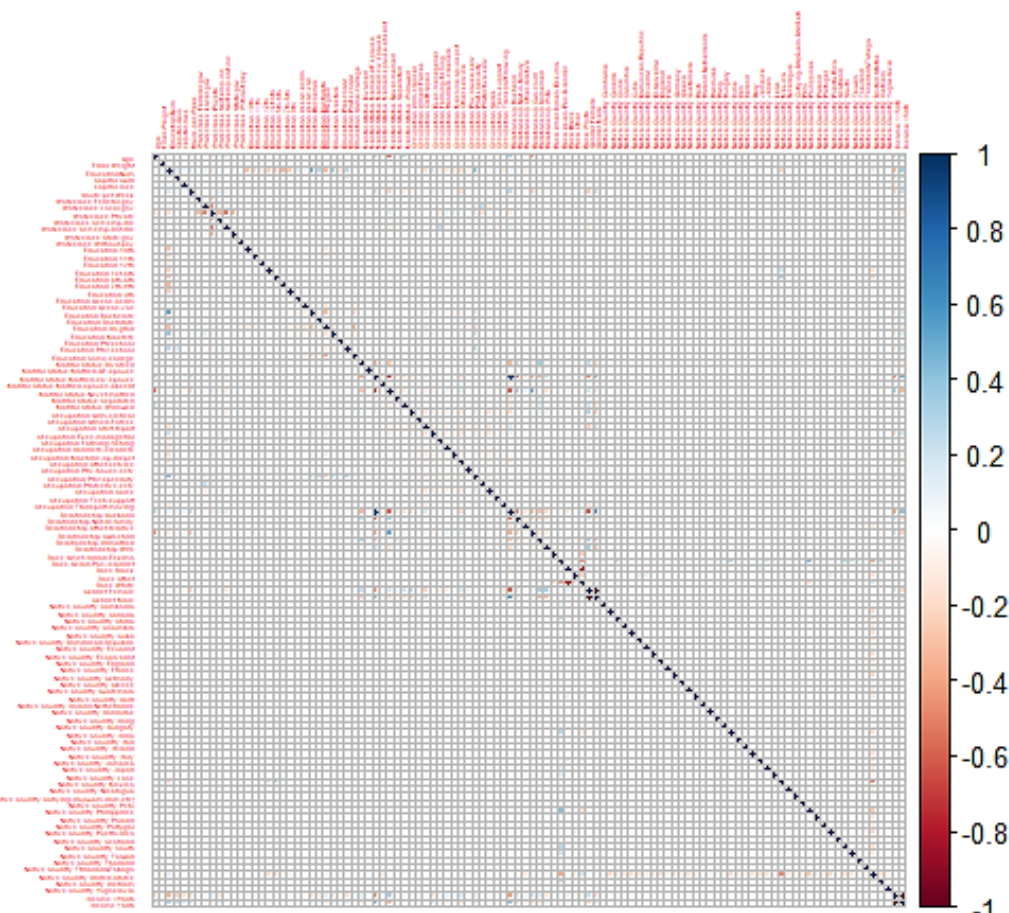
4. FEATURE SELECTION

Feature selection is the process of choosing a relevant variable out of many for the model building to improve the model's accuracy. The variable selection process must be done scientifically and not random as it directly affects the model's performance. The number of variables and the type of variables play a major role in reducing overfitting and underfitting inaccuracy in the model. This will lead to a model which is simpler and more accurate. We have done three types of variable selection which are Correlation Plot and Pair Plot Matrix, Backward Stepwise Regression, and PCA Plot. (Luna, 2021) (Mayureshrpalav, 2020) (Tripathi, 2019)

4.1 Correlation and Pair Matrix Plot

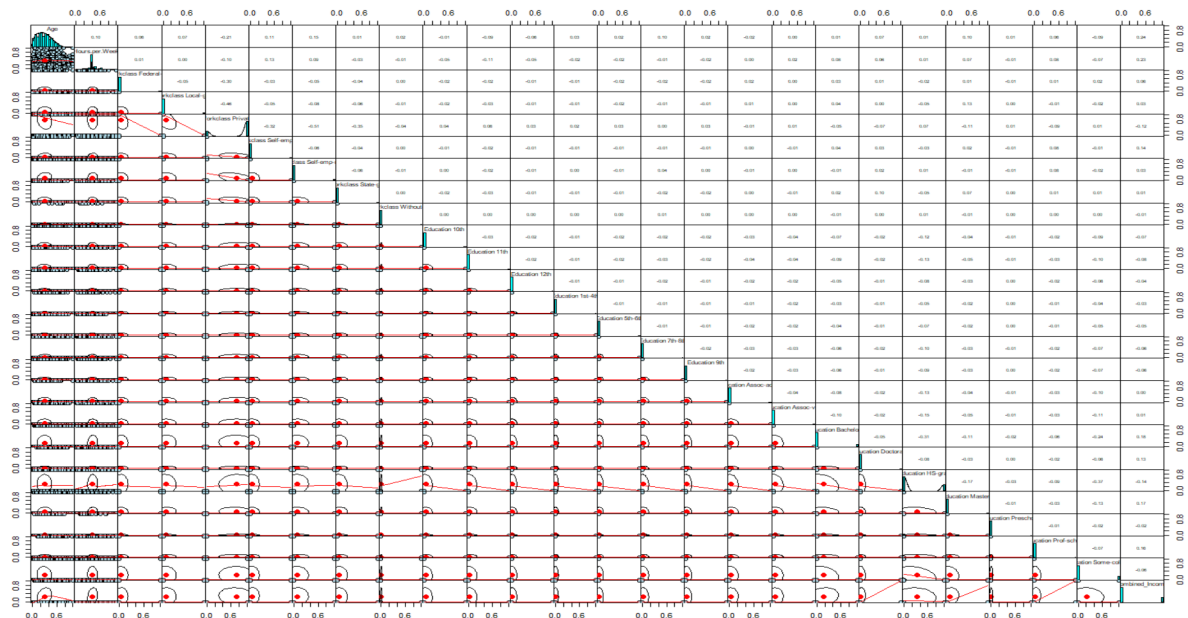
4.1.1. Correlation Plot

It is a graphical representation that shows the relation between multiple variables amongst them. It is a matrix format where the correlation indicates the strength and how correlated they are within them. This representation shows which variables are heavily correlated. According to the correlation plot we have selected the variables Age, and Hours. Per. Week, Work class, and Education.



4.1.2. Pair Plot Matrix (For selected variables - Age, Hours. Per. Week, Work class, and Education)

This plot visualizes the pairwise relationship between the variables, it graphically and numerically offers a comprehensive view of correlation, distribution, and weightage of each variable amongst the variables, where the correlation plot just shows the visual correlation coefficient of the variables amongst itself.



4.2. Backward Stepwise Regression

In this process of variable selection we begin with all independent variables included in the model. We sub-sequentially remove the variable that is of least significance based on statistical measurements such as p-values until a threshold is met. This helps the model to reduce the number of variables by showing the most impactful variables that contribute major change for the outcome or the dependent variable. This again as other methods help to reduce the model complexity but make sure the accuracy of the model is not affected. Based on this method the narrowed-down variables are - Age, Hours. Per. Week, Work class, and Education.

```
lm(formula = Combined_Income ~ Age + Final.Weight + EducationNum +
  Capital.Gain + capital.loss + Hours.per.week + `workclass Federal-gov` +
  `workclass Local-gov` + `workclass Private` + `workclass Self-emp-inc` +
  `workclass State-gov` + `Education 10th` + `Education 11th` +
  `Education 12th` + `Education 1st-4th` + `Education 5th-6th` +
  `Education 7th-8th` + `Education 9th` + `Education Assoc-acdm` +
  `Education Assoc-voc` + `Education Bachelors` + `Education Preschool` +
  `Marital.Status Divorced` + `Marital.Status Married-AF-spouse` +
  `Marital.Status Married-civ-spouse` + `Marital.Status Never-married` +
  `Occupation Adm-clerical` + `Occupation Craft-repair` + `Occupation Exec-managerial` +
  `Occupation Farming-fishing` + `Occupation Other-service` +
  `Occupation Priv-house-serv` + `Occupation Prof-specialty` +
  `Occupation Protective-serv` + `Occupation Sales` + `Occupation Tech-support` +
  `Relationship Husband` + `Relationship Not-in-family` + `Relationship Other-relative` +
  `Relationship own-child` + `Relationship Unmarried` + `Race Amer-Indian-Eskimo` +
  `Race Black` + `Race Other` + `Gender Female` + `Native.Country Cambodia` +
  `Native.Country China` + `Native.Country Columbia` + `Native.Country Greece` +
  `Native.Country India` + `Native.Country Mexico` + `Native.Country Outlying-US(Guam-USVI-etc)` +
  `Native.Country South` + `Native.Country Taiwan` + `Native.Country Vietnam`,
  data = normalized_data)
```

```
Residuals:
    Min       1Q   Median       3Q      Max
-0.99162 -0.25583 -0.02954  0.15181  1.25212
```

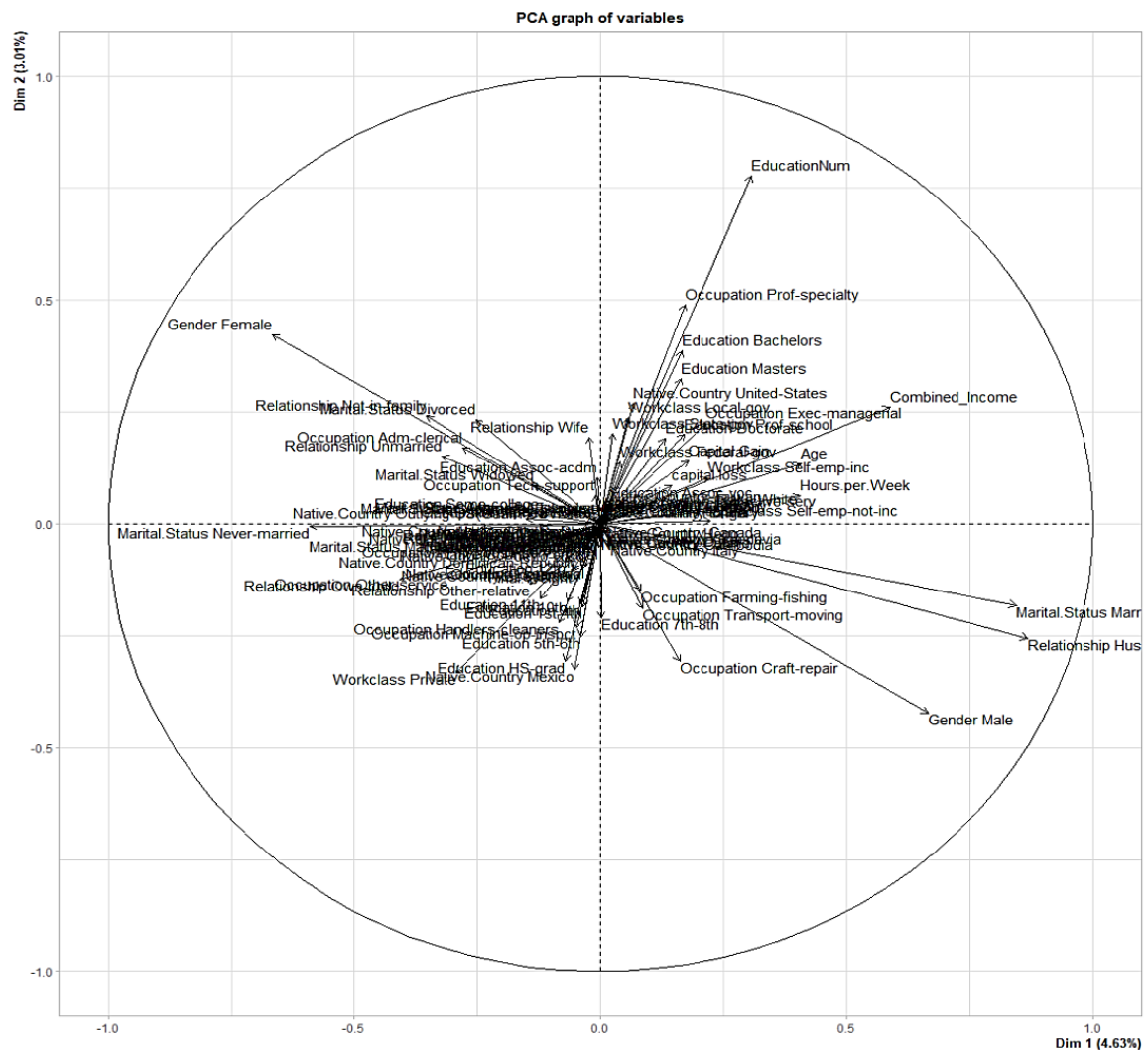
```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -0.359126   0.033593  -10.690 < 2e-16 ***
Age          0.206306   0.013904   14.838 < 2e-16 ***
Final.Weight 0.115590   0.028256    4.091 4.31e-05 ***
EducationNum 0.762937   0.025068   30.435 < 2e-16 ***
Capital.Gain 0.773157   0.027361   28.257 < 2e-16 ***
capital.loss 0.397795   0.021557   18.453 < 2e-16 ***
Hours.per.week 0.282622  0.017880   15.807 < 2e-16 ***
`workclass Federal-gov` 0.136326  0.013529   10.077 < 2e-16 ***
`workclass Local-gov`  0.034311  0.010817    3.172 0.001516 **
`workclass Private`    0.066921  0.007671    8.724 < 2e-16 ***
`workclass Self-emp-inc` 0.122165  0.012751    9.581 < 2e-16 ***
`workclass State-gov`  0.020467  0.012285    1.666 0.095705 .
`Education 10th`       0.113335  0.013782    8.223 < 2e-16 ***
`Education 11th`      0.092675  0.011918    7.776 7.70e-15 ***
`Education 12th`      0.063616  0.018194    3.496 0.000472 ***
```

**Image not of full value because of space constraint*

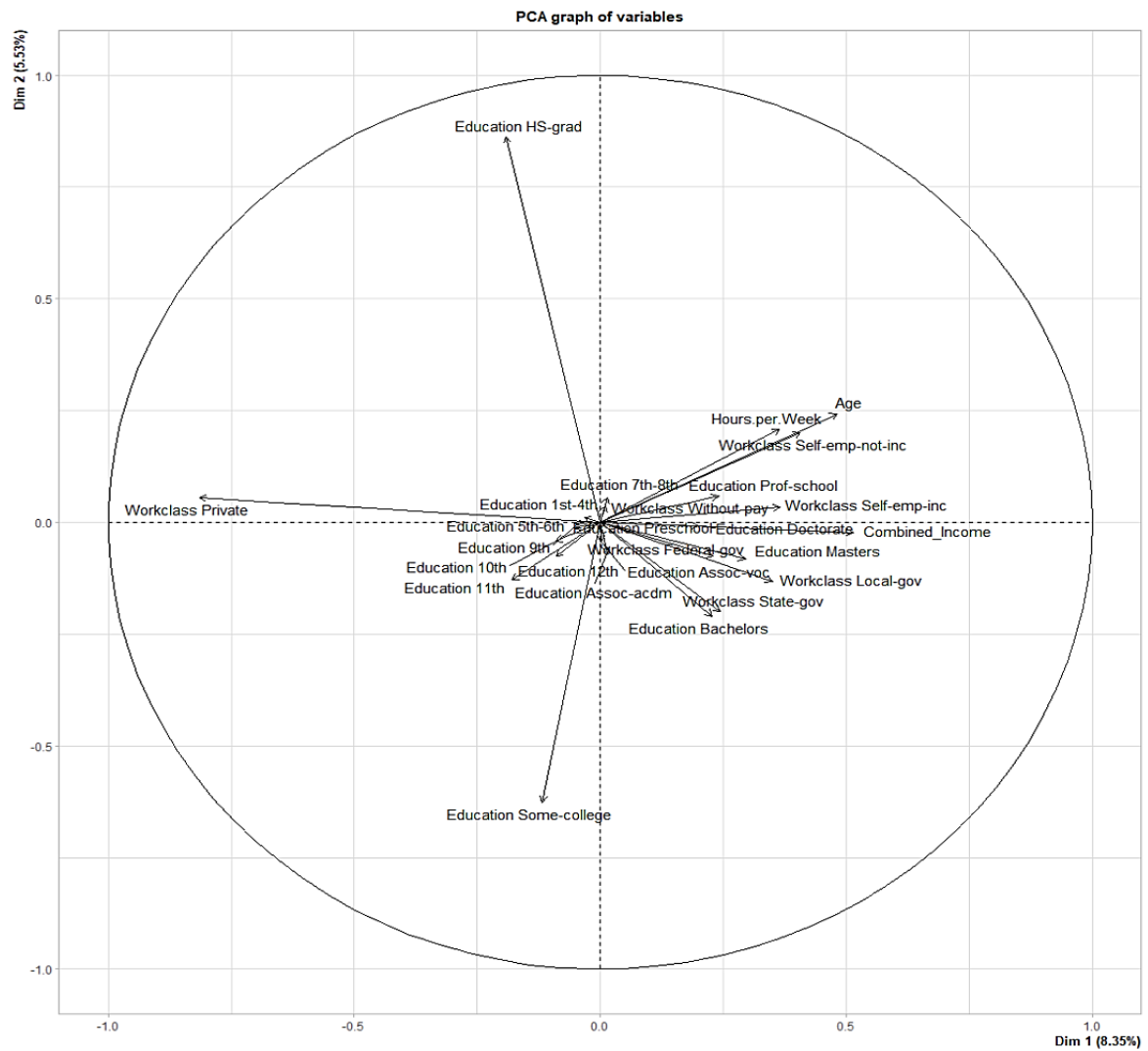
4.3. PCA (Principal Component Analysis)

It is a statistical technique that is mainly used to see which variables are on the same plane by doing this we can understand the importance of variables which can be later used to select the variables for the model. It transforms the variables into a set of orthogonal components called principal components which later shows the percentage of impact those variables have on the dependent variable. The PCA that we are using is a visual PCA which shows the similarity between variables by this visual representation we can select the variables that are not similar but are unique to each other.

4.3.1 PCA for all variables



4.3.2 PCA for selected variables - (Age, Hours. Per. Week, Work class, and Education)



5. ALGORITHM

Algorithms are step-by-step processes, formulas, or a set of rules that go in a sequence to solve a problem or perform tasks. In coding the algorithm serves as a framework on which the model processes data on how we tell it to execute, it does it efficiently. The use of correct algorithms is crucial based on the approach and data we use for the model to work at its best. (Hashemi-Pour, n.d.)

We will be using KNN Classification and the Naïve Bayes algorithm for this model. They both offer different approaches in their classification methods, which we can use to determine based on the result which is a better fit for this data's classification prediction model.

For training and testing, we will use the wide set approach of 80 percent and 20 percent of the whole data respectively. Using the 80-percentage data to train the data ensures the model will have sufficient data for the model to understand the relations for it to predict. With 20 percent of the remaining data, the model will be tested for its accuracy on the unseen and untrained data. This reduces the risk of overfitting and underfitting. Moreover, this is an industry standard that allows for consistent comparisons among models. (Gholamy, 2018)

6. KNN (K-Nearest Neighbours) CLASSIFICATION

KNN is a supervised machine learning classification algorithm that works on classifying the data points according to the majority of its nearby neighbors as per the given radius. This method does not use any underlying assumption of the data points but just solely classifies based on its neighbors. This approach of the KNN makes it very flexible and easy to grasp but its performance is highly dependent on the k value and the distance matrix. This method works exceptionally well in pattern recognition classifications. (Harrison, 2018)

6.1. Hyper Parameter Tuning

The KNN model is simple and effective, but the performance of the model mainly depends on the choice of the hyperparameter, which can be denoted by k. K represents the number of neighbors to be taken into consideration while making the prediction. Selecting an ideal k value is a crucial step in this process where the model's ability to understand the training data's neighbor and to implement the same to the unseen data. This will affect the model's accuracy, precision, recall, and overall strength.

The process of selecting the k value is a bit of an unorthodox process when compared to other systematic approaches, it is more like an experimental phase where we assign different values to k and choose the best choice based on its performance. Below are the reasons this step is a crucial factor for the model.

- Bias Variance Trade-off – The k value affects the model performance by overfitting or underfitting. A small k value can make the model capture noise from the training data which will result in overfitting. On the other hand, the k value with a higher value will lead to a more relaxed prediction which will make the model underfit with a high bias, hence we aim to find a right k that has a minimum bias and fitting issue.
- Model Stability – An effective k value ensures the model's stability and ensures that it performs consistently irrespective of the testing data set. This improves the model's ability to generalize certain neighbors so it performs as expected in new data which will ultimately be used for prediction.
- Performance Metrics – Each value of selected k will yield different performance metrics which are accuracy, precision, recall, and F1 score. These metrics help us to locate an ideal k value suitable for specific models based on the model's requirement. (Agrawal, 2023)

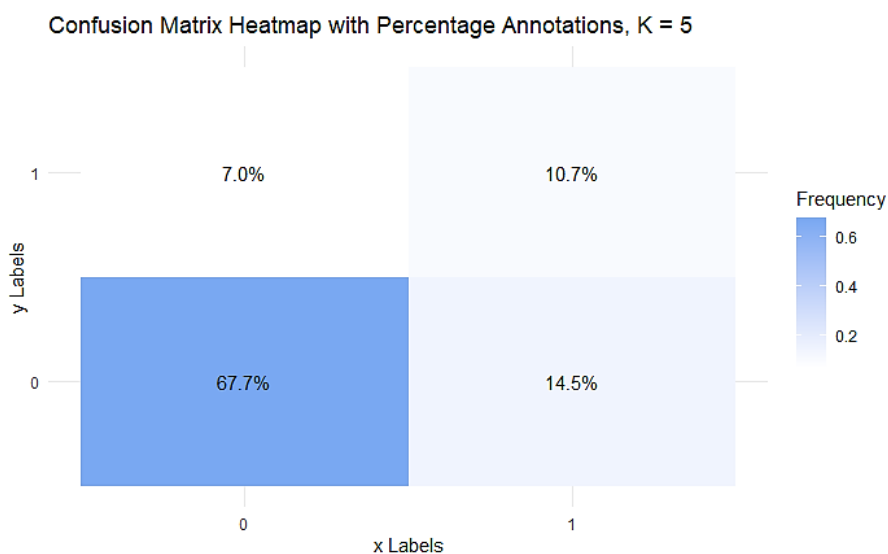
In our hyperparameter tuning, we selected k values 5, 7, and 9 to determine which amongst them is a better option by evaluating them using a confusion matrix. With metrics accuracy, recall, precision, and specificity.

6.1.1. Hyperparameter, k = 5

Cell Contents			
		N	
	N / Row Total		
	N / Col Total		
	N / Table Total		

Total Observations in Table: 6032

Test_Labels	Predicted_Test_labels_k5		Row Total
	0	1	
0	4079	431	4510
	0.904	0.096	0.748
	0.823	0.400	
	0.676	0.071	
1	876	646	1522
	0.576	0.424	0.252
	0.177	0.600	
	0.145	0.107	
Column Total		4955	1077
		0.821	0.179
			6032



True Negatives (TN)	4079
False Positives (FP)	431
False Negatives (FN)	876
True Positives (TP)	646
Total Observations	6032

Accuracy	78.38%
Recall	42.42%
Precision	60.02%
Specificity	90.44%
F1 Score	49.00%

The model with k =5 has an accuracy of 78.38%, with a precision of 60.02%, a recall of 42.42%, and an F1 score of 49%, this indicates a balanced performance between precision and recall for identifying positive cases.

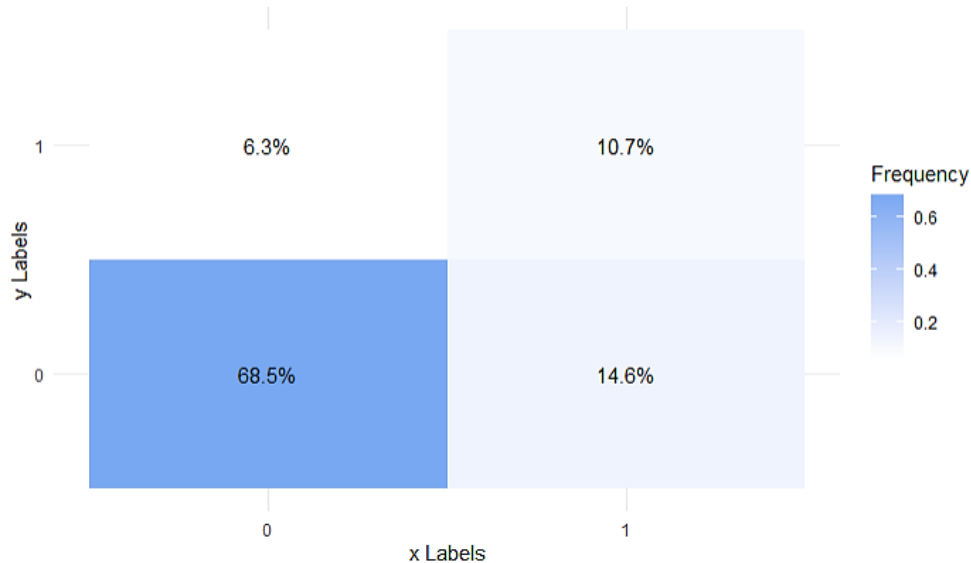
6.1.2. Hyperparameter, k = 7

Cell Contents			
			N
			N / Row Total
			N / Col Total
			N / Table Total

Total Observations in Table: 6032

Test_Labels	Predicted_Test_labels_k7		Row Total
	0	1	
0	4130	380	4510
	0.916	0.084	0.748
	0.824	0.372	
	0.685	0.063	
1	881	641	1522
	0.579	0.421	0.252
	0.176	0.628	
	0.146	0.106	
Column Total		5011	1021
		0.831	0.169
		6032	

Confusion Matrix Heatmap with Percentage Annotations, K = 7



True Negatives (TN)	4130
False Positives (FP)	380
False Negatives (FN)	881
True Positives (TP)	641
Total Observations	6032

Accuracy	79.09%
Recall	42.10%
Precision	62.80%
Specificity	91.59%
F1 Score	50.03%

The model with k =7 has an accuracy of 79.10%, with a precision of 62.8%, a recall of 42.10%, and an F1 score of 50.03%, this indicates a balanced performance between precision and recall for identifying positive cases.

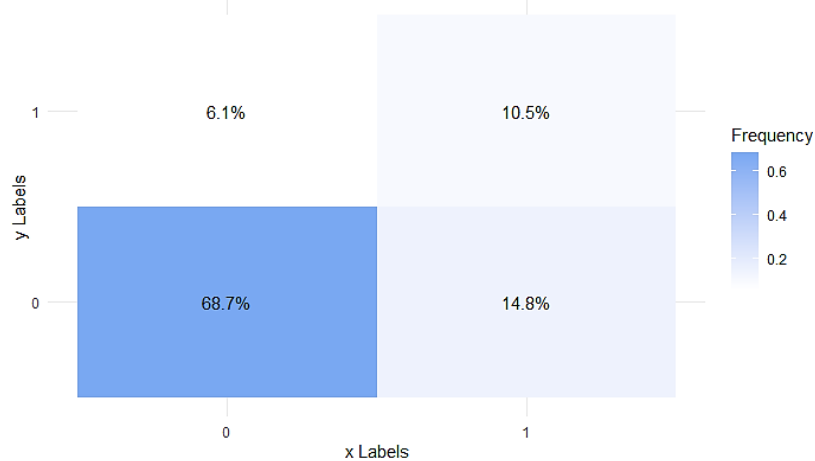
6.1.3. Hyperparameter, k = 9

Cell Contents	
	N
N / Row Total	
N / Col Total	
N / Table Total	

Total Observations in Table: 6032

Test_Labels	Predicted_Test_labels_k9		Row Total
	0	1	
0	4145	365	4510
	0.919	0.081	0.748
	0.823	0.366	
	0.687	0.061	
1	891	631	1522
	0.585	0.415	0.252
	0.177	0.634	
	0.148	0.105	
Column Total		5036	996
		0.835	0.165
		6032	

Confusion Matrix Heatmap with Percentage Annotations, K = 9



True Negatives (TN)	4145
False Positives (FP)	365
False Negatives (FN)	891
True Positives (TP)	631
Total Observations	6032

Accuracy	79.17%
Recall	41.50%
Precision	63.40%
Specificity	91.59%
F1 Score	50.10%

The model with k =9 has an accuracy of 79.17%, with a precision of 63.40%, a recall of 41.50%, and an F1 score of 50.10%, this indicates a balanced performance between precision and recall for identifying positive cases.

6.2. Conclusion

When compared to all the k values, k=9 had the highest scores. When compared to k=7 k =9 has better accuracy and improves the precision because of this we will go ahead with k=9 as a hyperparameter.

7. NAÏVE BAYES

Naive Bayes is a supervised machine learning algorithm that is used for classification models. The way this algorithm operates is that it assumes that all the variables are independent of each other and predicts based on probability where the data point should lie based on the variable's value or input. Naïve Bayes works well if the variables selected are good in correlation with the dependent variable, the drawbacks of this algorithm are if the variables are not that correlated and or are of lesser importance the prediction can be skewed. The assumptions of Naïve Bayes classification for this model are as follows,

- **Independence Assumption** - In our analysis, Naive Bayes assumes that a person's age, the number of hours they work per week, their work class (e.g. private sector, government employee), and their education level (e.g. high school diploma, Bachelor's degree) independently contribute to predicting their income level. The model will treat the probability of someone working a certain number of hours as completely unrelated to their age, work class, or education when it comes to their income. In reality, these variables might be related (for example, individuals with higher education might work more hours), but Naive Bayes ignores these potential relationships.
- **Likelihood of Features** - Naive Bayes will calculate the likelihood of each feature given each income class. For instance, it will estimate the probability that someone works over 40 hours a week given that they earn more than 50K and compare that to the probability of working the same hours for those earning 50K or less. Similarly, it will assess how likely someone with a Bachelor's degree is to earn more than 50K versus 50K or less. These probabilities are used to predict the income level for each individual in the dataset.
- **Feature Relevance** - In our analysis, Naive Bayes will assume that all of these factors—age, hours per week, work class, and education—are important in determining whether someone's income is above or below 50K. It assumes that knowing a person's work class is just as crucial for predicting their income as knowing their education level, even though in reality, some of these features might be more strongly related to income than others. It won't inherently weigh one factor more heavily than another unless the data suggests it based on the likelihood calculations. (Gamal, 2020)

7.1. Naïve Bayes evaluation metrics

For the purposes of evaluating the Naïve Bayes model, we will be using two metrics to see how well the model is performed. The two metrics that we will be using are the confusion matrix and the ROC (Receiver Operating Characteristic) curve.

7.1.1. Confusion Matrix

A confusion matrix is an accuracy measuring metric that is mainly used in classification models. It contains the details of the model's actual versus predicted values assigned by the model which allows us to check which instances were predicted accurately and which were inaccurately classified by the model.

Cell Contents	
	N
N / Row Total	
N / Col Total	
N / Table Total	

Total Observations in Table: 6032

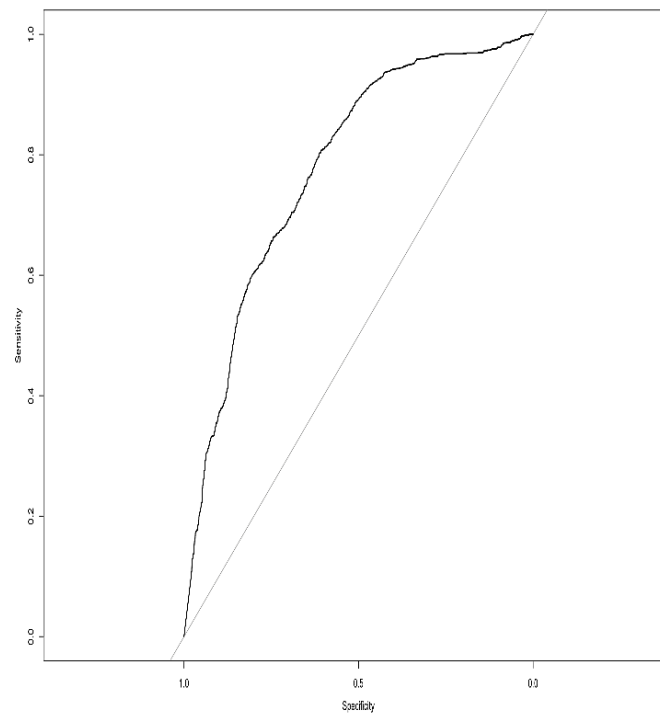
Test_NB\$Combined_Income	Predicted_Test_Labels_NB		Row Total
	0	1	
0	654	3903	4557
	0.144	0.856	0.755
	0.936	0.732	
	0.108	0.647	
1	45	1430	1475
	0.031	0.969	0.245
	0.064	0.268	
	0.007	0.237	
Column Total	699	5333	6032
	0.116	0.884	

True Negatives (TN)	654
False Positives (FP)	3903
False Negatives (FN)	45
True Positives (TP)	1430
Total Observations	6032

Accuracy	34.53%
Recall	96.90%
Precision	26.80%
Specificity	26.80%
F1 Score	41.96%

7.1.2. ROC (Receiver Operating Characteristic) Curve

The ROC curve is a graphical plot that shows how well the prediction model is working when compared to random prediction. The diagonal straight line is assumed the values if predicted at random and the curve is the model's prediction. The values used for this curve are true positive rate and false positive rate.



The ROC curve indicates the Naïve Bayes classification model is performed well as it is above the diagonal line. The steep rise of value in the curve suggests that the model is good at identifying true positives. The curve in short shows that the Naïve Bayes prediction model is better than random prediction.

7.2. Conclusion

The Naïve Bayes model did not predict as expected because of the below reasons.

- High Recall but Low Precision – The model performed well in the recall at 96.90%, which shows that it's very accurate at identifying most positive cases. But the precision was low at 26.80% which indicates a high level of false positives.
- Low Specificity and Accuracy – With a very low specificity of 26.80% the model does not correctly identify the false cases. Accuracy is also very low at 34.53% which shows the overall predictions of the model were very poor.
- Balanced Performance Indicated by ROC Curve – The ROC Curve suggests the model is above average even with the limitations mentioned above using the confusion matrix.

8. CONCLUSION

For any machine learning multiple models behave differently, so to measure which model performs better than the other there is a set of metrics that can be compared against these models to find the model that fits best for the specific type of use and the data.

The evaluation method that we will be using is a confusion matrix which gives values such as accuracy, precision, recall, and specificity along with F1 scores. We will not only take into account these metrics but also how simple the model is and the complexity of the model if there are any trade-offs.

Below is the comparison of KNN classification with $k = 9$ and Naïve Bayes classification models.

KNN

Cell Contents

		N
N / Row Total		
N / Col Total		
N / Table Total		

Total Observations in Table: 6032

Test_Labels	Predicted_Test_Labels_k9	0	1	Row Total
0		4145 0.919 0.823 0.687	365 0.081 0.366 0.061	4510 0.748
1		891 0.585 0.177 0.148	631 0.415 0.634 0.105	1522 0.252
Column Total		5036 0.835	996 0.165	6032

Naive Bayes

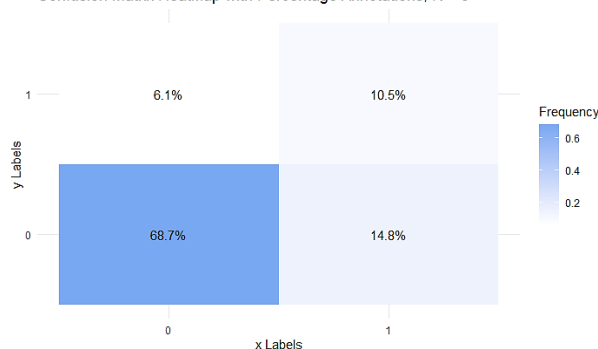
Cell Contents

		N
N / Row Total		
N / Col Total		
N / Table Total		

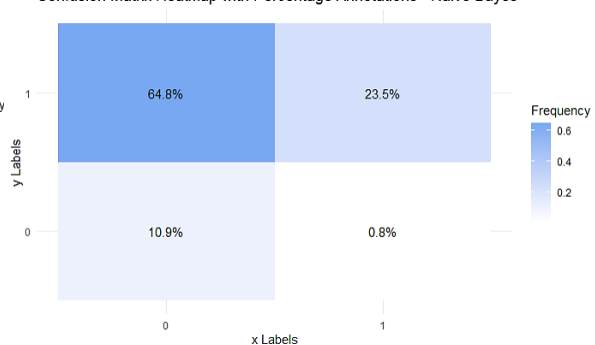
Total Observations in Table: 6032

Test_NB\$Combined_Income	Predicted_Test_Labels_NB	0	1	Row Total
0		654 0.144 0.936 0.108	3903 0.856 0.732 0.647	4557 0.755
1		45 0.031 0.064 0.007	1430 0.969 0.268 0.237	1475 0.245
Column Total		699 0.116	5333 0.884	6032

Confusion Matrix Heatmap with Percentage Annotations, K = 9



Confusion Matrix Heatmap with Percentage Annotations - Naive Bayes



KNN

True Negatives (TN)	4145
False Positives (FP)	365
False Negatives (FN)	891
True Positives (TP)	631
Total Observations	6032

Accuracy	79.17%
Recall	41.50%
Precision	63.40%
Specificity	91.59%
F1 Score	50.10%

Naïve Bayes

True Negatives (TN)	654
False Positives (FP)	3903
False Negatives (FN)	45
True Positives (TP)	1430
Total Observations	6032

Accuracy	34.53%
Recall	96.90%
Precision	26.80%
Specificity	26.80%
F1 Score	41.96%

From the above comparison, the KNN model performs better than Naïve Bayes in this specific scenario, with higher accuracy and F1 score. KNN is not only a superior model in this instance but also the simpler of both models. Hence, we would like to recommend the KNN classification model for this specific prediction project and dataset.

9. ETHICAL CONCERNS

Ethical concerns in prediction models have several critical considerations. From how the data is sourced to how the model predicts on few scenarios. A lot of care and tuning is to be done for the model so that there are no ethical concerns present. Below are the types of ethical concerns.

- **Fairness and Bias**
While training the model we must ensure that that the model does not unfairly discriminate any specific group regarding sensitive attributes such as gender or race. Any biases in the training can lead to biased prediction.
- **Transparency**
The model should be very transparent in the approach, methods, and workings of its prediction. This will help the stakeholders of the model to be informed about any possibility of bias, which can arise because of any steps or approaches taken.
- **Data Privacy**
The data collected and used for the model must be handled and dealt with very sensitively, making sure that if the information given by the interviewees had to be anonymous the information must remain confidential. If the information they provided was for one specific study or model, we must not store their information or use it for a different model.
- **Informed Consent**
We must ensure that the data given by the individuals are of their consent for use for a specified purpose.
- **Accountability**
Creating a prediction model will sometimes perform as not intended in the future for instances such as which are unintended or politically correct outcomes. We must monitor this over time and make necessary changes to the model.
- **Benefit vs Harm**
Do the prediction models' pros outweigh the cons such as stereotype reinforcing or aggravating inequalities? (Writer, 2024)

Our model's only ethical dilemma arises from data, the data was sourced from Kaggle, and they have free usage rights on the data, with consent from the individuals because of which our model is not impacted with ethical concerns.

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